

Section I: Data Structures [15 Marks]

Q 1) Create a random array of n integers. Accept a value x from user and use linear search algorithm to check whether the number is present in the array or not and output the position if the number is present. i) Accept & Display Function ii) Linear Search Function

Answer:

```
#include <stdio.h>
#include <stdlib.h>

void accept(int a[], int n)
{
    int i;

    for(i = 0; i < n; i++)
        a[i] = rand() % 100;
}

void display(int a[], int n)
{
    int i;

    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
}

int linearSearch(int a[], int n, int x)
{
    int i;

    for(i = 0; i < n; i++)
    {
        if(a[i] == x)
            return i + 1;
    }
    return -1;
}

int main()
{
    int a[100], n, x, pos;
```

```

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
        return 0;
    }

    accept(a, n);
    printf("Array: ");    display(a,
n);

    printf("Enter value x to search: ");
    scanf("%d", &x);

    pos = linearSearch(a, n, x);

    if(pos != -1)        printf("Number found at
position %d", pos);    else        printf("Number
not found");

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Create the following tables (primary keys are underlined).

Person (pnumber, pname, birthdate, income),

Area(aname,area_type).

There exists a one-to-many relationship between Area and Person.

Constraint: ‘area_type’ should have value either urban or rural.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Area(
    aname VARCHAR(50) PRIMARY KEY,
    area_type VARCHAR(10)
        CHECK(area_type IN ('urban','rural'))
);

```

```
CREATE TABLE Person(      pnumber INT PRIMARY
KEY,      pname VARCHAR(50),      birthdate DATE,
income NUMERIC(10,2),      aname VARCHAR(50)
REFERENCES Area(aname)
);
```

```
INSERT INTO Area VALUES
('Pune','urban'),
('Delhi','urban'), ('Wadgaon','rural');
```

```
INSERT INTO Person VALUES
(1,'Ravi','1990-02-15',30000,'Pune'),
(2,'Sita','1992-03-10',8000,'Wadgaon'),
(3,'Ram','1988-02-21',30000,'Delhi');
```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. List the details of persons with income greater than 10000.

Answer:

```
SELECT * FROM Person WHERE income >
10000; ii. Transfer all people living in 'Pune' to
'Delhi'.
```

Answer:

```
UPDATE Person
SET aname = 'Delhi'
WHERE aname = 'Pune';
```

iii. Count number of area's which has type 'urban'.

Answer:

```
SELECT COUNT(*) FROM Area WHERE
area_type = 'urban';
```

iv. Give the count of people who are born in the month of February.

Q 2(C)

C. Write a query that count the total number of people staying in “urban” area with income equal to average income. [4 marks]

Answer:

```
SELECT COUNT(*)  
FROM Person P  
JOIN Area A ON P.aname = A.aname  
WHERE A.area_type = 'urban'  
AND P.income = (SELECT AVG(income) FROM Person);
```

Q 3) Viva [5 marks]